

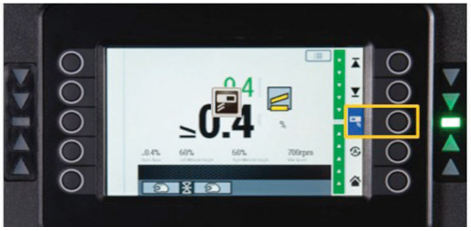
Preparing to Pave (Slope)

When preparing to pave under slope control, set up the paver and screed using Paving by the Numbers. Once this has been completed, touch the Sensor Select icon on the Guidance screen. Choose the slope sensor icon and press and release the “OK” soft key.

NOTE: The Display unlock button on the screed operator station key pad must be selected in order to utilize the touch screen function.

NOTE: Only one side can be selected for slope at a given time.

Preparing to Pave (Slope)



Press and hold the Bench soft key until an audible beep or until the bench icon appears on the Guidance screen, this benches the angle of the screed.



Press and release the Manual / Automatic soft key to place the system into the Automatic mode. This will allow the slope control system to control the tow point cylinder for that side actuated for slope. The control system will then control the angle of the screed when the paving parameters have been met.

If the screed angle needs to be adjusted, press the arrow soft keys up or down to make the required correction.



Preparing to Paving (Slope) ***(Continued)***

If the guidance number does not match the slope of the mat being laid, the slope sensor will need to be calibrated. To do this, touch the Measured Data area on the Guidance screen. A key pad will appear on the screen. Touch the correct percent slope of the measured mat and also verify that the slope direction is correct. When the correct number and slope direction is entered press and release the “OK” soft key to lock in the calibrated number. The slope sensor is now calibrated and should match the mat being laid.

Auto Valve Calibration



Next, we will walk through the Auto Valve Calibration on the F-series Grade and Slope system. When the machine first gets the system installed a calibration must be done to store the characteristics of the tow point valves. The Grade control system applies electrical current to the tow point valves in order to determine the operating range of the valve. The operating range of the valve is critical in order to have a properly performing Grade control system. The calibration values will then be stored in the electronic control system so if a display is replaced a calibration will not be needed as those calibration values will be download upon power up of the machine and system. Caterpillar recommends preforming the calibration periodically for optimum system performance. This calibration must be done when the tow point valves, cylinders or the hydraulic system has changed in someway.



Auto Valve Calibration *(Continued)*

NOTE: The left and right valve calibrations are identical. Repeat this procedure for both sides of the machine.

NOTE: Calibration can only be done on one side at a time. Once the calibration has been started on the display do not interact with the Grade and Slope screens on the other displays or actuate any switches.

Auto Valve Calibration - Setup

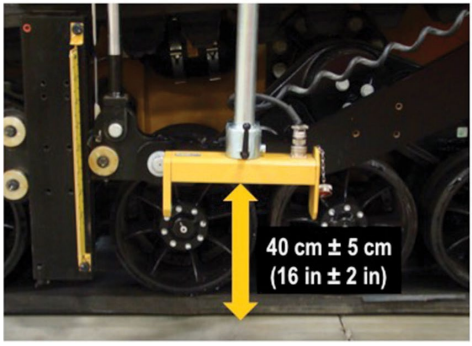


Occasionally there is a need to calibrate the F-series Cat Grade and Slope system, this task can be achieved very easily.

To start, move the machine to a flat surface. Then position each tow point cylinder so it has equal stroke up and down. Lower the screed by pressing the "Float" button on the instrument panel when in the **"Pave mode"**.

Slowly move the machine forward to remove any slack in the tow point assembly.

Next, null the screed so no tension can be felt in the thickness control screws.



Position the Sonic Tracer slightly behind the tow point pin. (As shown)

Position the Sonic Tracer as close to the machine as practical and without sensing the edge of the track.

Position the height of the Sonic Tracer 40 cm ± 5 cm (16 in ± 2 in) above the reference.



Next setup the paver controls so the system is active and it can be calibrated.

To do this:

1. Apply the parking brake.
2. Place the Propulsion switch in the Pave mode
3. Place the screed into “Float”
4. Engine rpm to low idle
5. Squeeze the trigger on the propulsion lever and move to the full forward position

NOTE: Screed must be in “Float” or the system will NOT allow for the automatic calibration to be achieved.

The system hydraulics temperature must be at least 38º C (100º F), but normal operating temperature is preferred.





The ST200 is the only sensor that can be used for calibration. Touch the sensor select icon on the Guidance screen (1). Select the ST200 sensor icon for the “Surface mode” (2). Press and release the “OK” soft key (3).

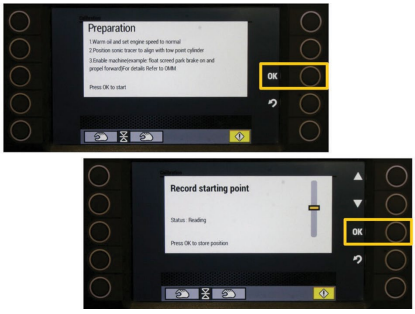


The machine is now set up in the proper position and the proper state to allow for final completion of the automatic calibration. Proceed to the Guidance screen. In the upper right hand corner touch the System menu icon (1). In this menu touch the Installation menu (2) choice.

NOTE: Going into the Installation Menu at anytime, when in "Automatic," will place the system into the "Manual mode".

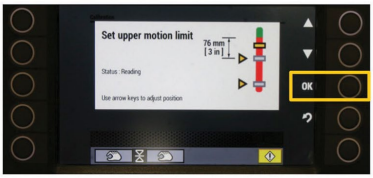


At this time, choose the Calibration menu. The choices will be Left or Right Valve Menu's. Choose the side being calibrated by touching the screen or by highlighting and pressing the "OK" soft key, then proceed to the next steps and follow the directions.

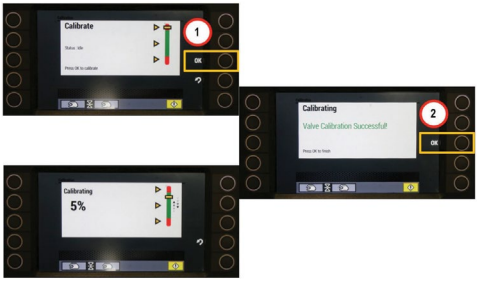


Since the machine was set up in prior steps (pages 79-85), the only step that will need to be completed when the "Preparation" menu appears is to press and release the "OK" soft key. The next screen will instruct the operator to store the position of the tow point cylinder. The cylinder should be positioned properly already but if not move cylinder to the middle of the cylinder stroke and press the "OK" soft key to store the middle position.





The next screen to appear is the lower motion limit. To achieve this step the operator must press the “down” arrow soft key to move the tow point cylinder approximately 76-100 mm (3-4 in) down from the middle position. Once this position is accomplished the yellow bar will move into the green zone. Note: If it is not in the green zone you will NOT be able to store that position. When in the green area, Press and release the “OK” soft key. The next screen will be for the upper motion limit. Press the “up” arrow soft key to move the tow point cylinder up until the yellow bar moves into the green zone on the screen. This position will be 76-100 mm (3-4 in) above the middle position. Again, once in the green zone press and release the “OK” soft key.



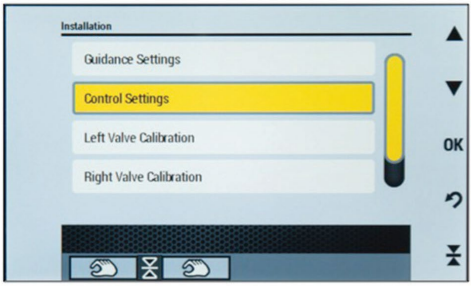
Once the positions have been stored, press the “OK” soft key to begin the calibration process. The system will automatically move the tow point cylinder locking in the proper currents to achieve the proper flow and speed of the tow point cylinder. This needs to be done on both sides of the machine because each valve has different characteristics. Once the system has completed the calibration process the screen will display “Valve Calibration Successful!”. At this time the operator must press and release the “OK” soft key to achieve the calibration.





After the setup is complete, press and release the "Play" soft key. At this time the grade control system will automatically calibrate to achieve the proper system responses required to control the machine tow point valves. DO NOT activate any switches on the screed during the calibration process or the system will "abort" the calibration.

Once the screen displays "Valve Calibration Successful" the operator must press the "OK" soft key to complete the calibration.



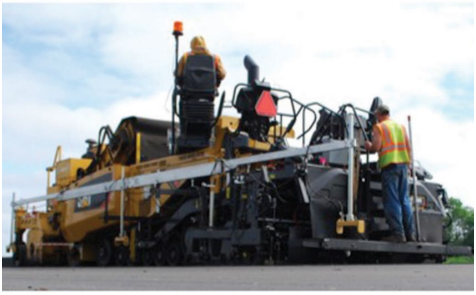
If you run into a situation where you need to reset the system back to the factory defaults follow these steps.

Under Installation menu is the Control Settings menu. This menu allows for configuration of the Valve Speeds, the On-Grade Deadbands, Elevation Control Windows, Cross Coupling and Reset to Defaults submenus.



The Reset menu allows the operator to set all "Control Settings" back to default numbers. Resetting is recommended as a starting point if there are undesirable control characteristics.

NOTE: Resetting the default values will not reset the valve calibration values.



For further information on the Grade and Slope system for the “F” Series Asphalt Pavers please refer to the Operations and Maintenance Manual for your machine You can also refer to the handbook for the Grade and Slope system for the “F” Series Asphalt Pavers QEDQ1959.



AVERAGING SKIS ARE KEY TO SMOOTHNESS

The majority of state Departments of Transportation (DOT) now have payment incentives and penalties written into their specifications for pavement smoothness. Whether the specification is written around profilograph results or, more recently, around the International Roughness Index (IRI), project profitably may hinge on your ability to collect a significant amount of the available smoothness incentive.

It is now absolutely clear that proper use of averaging skis on ride incentive projects is the key to maximizing the amounts of incentives earned. While doing the fundamental things right, usually called

best paving practices, is still very important, the inclusion of averaging skis into standard highway paving practice has become a requirement

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Skis come in various lengths and configurations. Some reference a target outside the paving width. Some reference inside the paving width. Some even extend over the screed. When selecting the right ski (mobile grade reference), two factors should always be considered first: (1.) the longer the better and (2.) the more articulation points the better.

A variety of averaging devices are available from Caterpillar and other paver manufacturers. Following is a review of the Caterpillar averaging skis.

Tips for Using Skis

1. Install any averaging device so the center of the string or beam is just behind the tow point cylinder. This is the ideal point for grade sensor location when ultimate smoothness is the goal. Remember, on a device like the Outboard Leveler, the center of the string will show the most reduction in grade deviation. The tow point reaction will be exactly the same amount as the sensed grade deviation and mat smoothness will be the best when the center of the string or beam is in this position.

2. Position the grade sensor (mechanical or sonic) in the middle of the string. Adjust the tension of a mechanical sensor arm so the arm maintains good contact with the string. Position sonic sensors 40 cm ± 5 cm (16 in ± 2 in) above the string.
3. Adjust string tension so there is no slack or vibration. Use string that is at least 3 mm (1/8 in) in diameter to provide a good target for sonic sensors. String must be a minimum of 10 cm (4 in) above beam.
4. Do daily maintenance on towed skis. The feet that contact the grade can pick up tack. Excessive tack build-up can cause the towed ski to vibrate excessively. Clean the ski feet at the end of each shift.

Using Two Skis

On any multi-lift project, install averaging skis on both sides of the paver for every lift after the bottom lift. On the bottom lift, it is common to operate with an averaging device on the centerline side of the paver and slope control for the shoulder side.

This will build the correct profile as specified in the project plan. If the profile of the structure is already correct, as is the case on some overlay or mill/fill projects, operate with two skis on every lift.

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On smoothness projects, once the correct profile is built, your main concern is mat smoothness. So, look at the grade reference possibilities. Determine whether you can sense a reference outside the paving width or if you have to sense inside the paving width. Then, select the type of averaging device that best matches the paving application. Remember, the longer the better and the more articulation points the better.

Do not hesitate to use an averaging ski when matching a longitudinal joint. In the past, the conventional practice was to use a single sensor directly referencing the adjacent mat for joint match. A single sensor without an averaging ski produces a fast reacting screed, good for exact joint match but bad for smoothness.

Ride incentive money is out there. You should be earning as much of it as possible on every project. Put averaging skis to work for you. You'll see better results.



OUTBOARD LEVELER

The Outboard Leveler is towed by the paver and references the grade outside the paving width. I comes in two lengths: 9.15 m (30') or 12.2 m (40'). It has multiple feet and multiple articulation points. It has the ability to reduce the amount of sensed grade deviation by a factor of eight. In other words, if a ski foot runs over a 25 mm (1") deviation, the center of the string running between the ends of the ski will move only 3 mm (1/8"). The grade sensor (either sonic or mechanical) will send a signal to the control box signifying a 3 mm (1/8") correction is required. The screed will make a small, slow correction and the deviation will be spread out over a long portion of the mat.

The Outboard Leveler is judged the ultimate averaging device for smoothness. It is recommended for highway projects when long pulls are the norm. Its main drawback, when compared to non-contacting averaging skis, is that it is harder to transport because of its size and number of moving parts.



FORE AND AFT LEVELER

The Fore and Aft Leveler mounts inboard of the paving width and bridges the screed extender to reference ahead of the screed and behind the screed. The front section is similar to the multi-articulated portion of the Outboard Leveler and the rear section is a wide foot that is towed by the screed and rides on the freshly laid mat. One significant benefit of the Fore and Aft Leveler is that the rear portion uses the smooth mat as a reference. This method is very popular when there is no suitable grade reference outside the paving width.

It is often used on the shoulder side of the paver in conjunction with the Outboard Leveler used on the centerline side of the paver.

It is used mainly on highway jobs with long continuous pulls. The rear foot must be heated prior to use on the hot mat. The rear portion blocks access to the outside of the screed extender.



SONIC AVERAGING BEAM

The Sonic Averaging Beam is the most recently developed averaging device. The beam is solidly attached to the paver and can be installed to sense the grade outboard or inboard of the paving width. Depending on the manufacture, the beam could be from 8.5 m to 9.1 m (28’ to 30’). The signals from the three or four sensors, depending on ski, are combined and averaged by the automatic grade control box. Its averaging capabilities are somewhat less than the Outboard Leveler. It has become very popular for a variety of reasons. Because it is attached to the paver and does not contact the grade reference, it is easy to transport compared to “drag” skis. Because it can sense grade both outboard and inboard of the paving width, it has the most versatility and can be used for nearly any application. Since it is non-contacting, it does not pick up tack and requires less maintenance.



CONCLUSION

This book covers a lot of different things that hopefully will improve your paving operation.

Some of the key points to remember are:

Follow Paving by the Numbers.

Constant paving speed – Calculated at the beginning of every job.

Constant head of material – Keep the feed sensors adjusted properly and the head of material consistent.

Screed adjustments – Make minor adjustments giving the screed time to react. Remember the basics. It takes five tow arm lengths to complete a change made to the screed.

Grade sensor position – Know what you want to do and the type of reaction you need from the sensor.